

1.1 Vectors and Linear Combinations

begin: { Calculus: $x, f(x)$
Linear algebra: vector, linear combination.

task 1: linear combination $cv + dw$

describe all combinations e.g. plane / line

Can / can't find c, d for a specific vector.

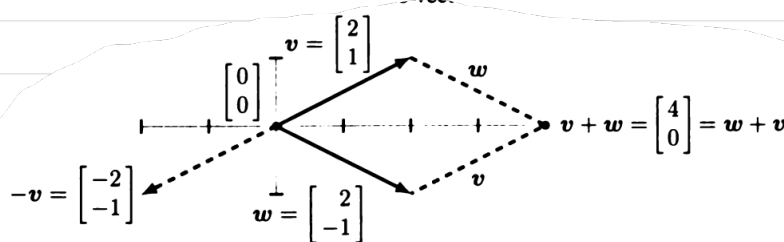
introduce more vectors in higher dimensions (n)

m rows n columns $m \times n$ matrix $A = [v_1 \ v_2 \ \dots \ v_n]$

task 2:

$\Rightarrow$ go back to v, w

task 1:



The vectors cv (for all numbers c) fill an infinitely long line in the xy plane. If w is not on that line, then the vectors dw fill a second line. We aim to see that the linear combinations $cv + dw$ fill the plane.

also: $c=d=0 \Rightarrow$ must go through zero vector.

col vector form

task 2: e.g. solve $c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix} \Rightarrow \begin{cases} 2c + 2d = 8 \\ c - d = 2 \end{cases}$

matrix form: $\begin{bmatrix} 2 & 2 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$

(1): 引出三种列式表达 linear combination 的方式

Column Way, Row Way, Matrix Way		
Column way Linear combination	$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + d \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$	
Row way Two equations for c and d	$v_1c + w_1d = b_1$ $v_2c + w_2d = b_2$	
Matrix way 2 by 2 matrix	$\begin{bmatrix} v_1 & w_1 \\ v_2 & w_2 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$	

就是由方程式/向量/矩阵外推定义的!

与乘的概念是统一的, 并可推广到 AB

下一节!

$\Rightarrow$ 就叫 A 乘 B 与矩阵的乘法.

(2) 何时/有解/能被 linear comb 表达

对于 two (1) independent vectors $\Rightarrow$ 有解 $\rightarrow$ 对于 2×2 矩阵.

(2) dependent $\Rightarrow$ 不-定

(3) $m < n \Rightarrow$ 可能无解; column space of A is a subspace of $\mathbb{R}^n$.

e.g. $\begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$

选择 $\leftarrow$

special matrix: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad Ax = x$